

Ubuntu Domain Join and Verification Documentation

Overview

This document outlines the steps taken to join an Ubuntu machine to an Active Directory (AD) domain using `realmd`, `sssd`, and related tools. The process includes DNS configuration, domain discovery, joining, user login verification, and automatic home directory creation.

1. DNS Configuration

Before joining the domain, the DNS settings were configured using `resolvectl` to ensure the domain controller could be resolved:

```
sudo resolvectl dns eth0 192.168.100.10
```

```
sudo resolvectl domain eth0 stellar.local
```

The `ping` command initially failed until Avahi (`.local` mDNS) was disabled and the configuration was refreshed.

2. Disable Avahi Daemon

To prevent `.local` conflicts with mDNS, Avahi was disabled:

```
sudo systemctl disable avahi-daemon
```

```
sudo systemctl stop avahi-daemon
```

3. Join Domain

The system was joined to the domain using the `realm` command. A successful join was verified with `realm list`:

```
sudo realm join --user=administrator stellar.local
```

```
realm list
```

4. Allow Domain Logins

Domain users were allowed to log in via:

```
sudo realm permit --all
```

5. Domain User Login

After joining, the user 'jdoe@stellar.local' was able to log in from the GUI and the home directory was automatically created upon first login. This confirms successful domain integration.

6. Verification

Commands used to verify domain user login and DNS resolution:

```
id
```

```
echo $HOME
```

```
ping stellar.local
```

```
ping 192.168.100.10
```

Title: Joining Ubuntu to Active Directory Domain with DNS Resolution Fix and Login Verification

Objective: To join an Ubuntu 22.04 VM to a Windows Active Directory (AD) domain, ensure DNS resolution works, verify domain login, and confirm automatic home directory creation for domain users.

Environment Details:

- **Ubuntu VM hostname:** stellarubuntu
 - **Domain Controller (DC):** Windows Server 2022
 - **Domain name:** stellar.local
 - **Domain user:** jdoe@stellar.local
 - **DNS server (DC IP):** 192.168.100.10
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Step-by-Step Instructions:

1. Configure DNS Resolution

Command:

```
sudo resolvectl dns eth0 192.168.100.10
```

```
sudo resolvectl domain eth0 stellar.local
```

Verify with:

```
resolvectl status
```

Make sure it shows the DNS server and domain under the correct interface.

Test DNS name resolution:

```
dig stellar.local
```

```
nslookup stellar.local
```

If ping to stellar.local fails even though dig/nslookup works:

2. Disable Avahi and Correct NSS Order

Avahi may interfere with .local domains. Disable it:

```
sudo systemctl disable avahi-daemon
```

```
sudo systemctl stop avahi-daemon
```

Edit the NSS configuration:

```
sudo nano /etc/nsswitch.conf
```

Find the line:

```
hosts: files mdns4_minimal [NOTFOUND=return] dns
```

Change it to:

```
hosts: files dns
```

Then reboot the machine:

```
sudo reboot
```

Verify ping to DC:

```
ping stellar.local
```

```
ping 192.168.100.10
```

3. Install Required Packages

```
sudo apt update
```

```
sudo apt install realmd sssd sssd-tools libnss-sss libpam-sss adcli samba-common-bin -y
```

4. Discover and Join the Domain

```
realm discover stellar.local
```

```
sudo realm join --user=administrator stellar.local
```

Provide the AD password when prompted. If successful, confirm with:

```
realm list
```

You should see details like:

- configured: kerberos-member
- server-software: active-directory

- client-software: sssd
-

5. Allow All Domain Users to Log In

```
sudo realm permit --all
```

To limit to a specific user:

```
sudo realm permit jdoe@stellar.local
```

6. Enable Automatic Home Directory Creation

Edit or create the following file:

```
sudo nano /etc/pam.d/common-session
```

Add this line at the end:

```
session required pam_mkhomedir.so skel=/etc/skel/ umask=0022
```

7. Log Out and Log In as Domain User

Reboot or log out. On the login screen, you should now see domain users listed (e.g., John Doe).

If not visible, use "Not listed?" and enter:

```
jdoe@stellar.local
```

After successful login, verify:

```
echo $HOME
```

```
id
```

```
ls /home
```

Confirm /home/jdoe is created.

Test network connectivity:

```
ping stellar.local
```

```
ping 192.168.100.10
```

Outcome:

- Ubuntu VM is joined to the AD domain.
 - DNS resolution works correctly.
 - Domain user jdoe@stellar.local can log in.
 - Home directory /home/jdoe is auto-created on first login.
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